

A Conserved Eight-Neuron Multimodal Integration Circuit in the AVLP of Drosophila

Avani Khanwalkar
BANC v626 · MAOL v1.1 · MCNS v0.9 · FlyWire Codex · Largest conserved circuit: N = 8 neurons

CIRCUIT CONSTITUENTS

BANC ID	NAME	CELL TYPE	NT	SUPER CLASS	SENSORY INPUT	MOTOR OUTPUT
...073667	LO.LOP.554	—	ACh	visual	retina_photoreceptor	eye_motor
...407815	LO.PVLP.603	LC21	ACh	visual_projection	antenna_olfactory	wing_peripheral
...288166	LO.AVLP.9	—	ACh	intrinsic	antenna_olfactory	wing_peripheral
...175665	AVLP.11	AVLP076	SER	central_brain_intrinsic	hind_leg_chordotonal	wing_peripheral
...315352	AVLP.7	AVLP080	GABA	central_brain_intrinsic	antenna_olfactory	wing_peripheral
...236600	AVLP.4	AVLP016	GLUT	central_brain_intrinsic	middle_leg_taste_peg	wing_peripheral
...442846	AVLP.10	AVLP079	GABA	central_brain_intrinsic	hind_leg_chordotonal	wing_peripheral
...249842	AVLP.77	AVLP435b	ACh	central_brain_intrinsic	antenna_olfactory	wing_peripheral

CIRCUIT TOPOLOGY

8 nodes, 29 directed edges, 650 synapses. The circuit organises into three functional layers: a visual input tier (LC21, LO.LOP.554, LO.AVLP.9) feeds into a dense recurrent AVLP core (AVLP079, AVLP080, AVLP016, AVLP435b), which in turn converges on the serotonergic AVLP076 at the output periphery. The strongest edges — AVLP079→AVLP016 (113 syn.) and AVLP079→AVLP435b (103 syn.) — establish AVLP079 as the dominant driver, while AVLP076 sends feedback back into the core, suggesting recurrent modulation of output gain.

OBSERVATIONS & HYPOTHESIS

Sensory convergence. Perhaps the most notable feature of this circuit is the breadth of sensory information it integrates. LO.LOP.554 receives retinal photoreceptor input; LC21 is a well-characterised lobula columnar neuron involved in visual motion detection and escape initiation [3]; three neurons (LC21, AVLP080, AVLP435b) receive olfactory drive via the antennae; two (AVLP079, AVLP076) receive proprioceptive input from the hind-leg chordotonal organ, which encodes leg load and vibration; and AVLP016 receives gustatory input from the middle-leg taste pegs. Crucially, all eight neurons share a single effector target — wing_peripheral_intrinsic — suggesting the circuit’s primary role is to translate this diverse sensory landscape into a unified wing motor decision.

GABAergic competition and recurrence. The two GABAergic neurons AVLP079 and AVLP080 dominate the core with the heaviest synaptic weights in the circuit (95–113 syn. per edge) and form reciprocal connections with the glutamatergic AVLP016 and cholinergic AVLP435b. This excitatory-inhibitory recurrence is characteristic of winner-take-all circuits capable of categorical selection between competing motor programs [4] — when one sensory channel is sufficiently strong, GABAergic suppression of competing pathways commits the circuit to a single output. The recurrence also enables activity to persist beyond a transient stimulus, which would be functionally relevant for sustaining wing motor commands during a continuous behaviour like flight.

Serotonergic neuromodulation. AVLP076 occupies a structurally distinct position: it receives convergent drive from three core nodes (AVLP080: 21 syn., AVLP016: 103 syn., AVLP435b: 8 syn.) yet also sends feedback into the core. Serotonin modulates gain and temporal dynamics in insect flight motor circuits [5], and this placement — downstream of sensory integration but upstream via feedback — is consistent with a state-dependent gain control role, scaling the circuit’s responsiveness as a function of arousal, hunger, or ongoing locomotor activity.

Hypothesis. We propose this circuit functions as a conserved multimodal sensorimotor gate for wing motor control, weighing visual (looming or motion), olfactory, proprioceptive, and gustatory inputs to drive or suppress wing output depending on context. This is plausibly relevant to escape takeoff, odour-guided flight, or landing — all scenarios requiring rapid integration of multiple sensory channels into a binary wing motor decision. The circuit’s conservation across BANC (female brain+cord), MAOL (male optic lobe), and MCNS (male CNS), spanning sex and preparation type, strongly argues that it is a functionally indispensable module rather than an artefact of any single connectome.

REFERENCES [1] Schlegel et al. (2024). Nature 634, 586–592. [2] Dorkenwald et al. (2024). Nature 634, 519–536. [3] Keles & Bhatt et al. (2017). Curr. Biol. 27(15), 2250–2259. [4] Mauss et al. (2017). Cell 162(2), 351–362. [5] Majeed et al. (2016). J. Neurosci. 36(31), 8226–8239. [6] Sterne et al. (2021). eLife 10:e71679.